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Saveetha Institute of Medical And Technical Sciences
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**Data-Driven Public Transport Delay Analysis Using
Exploratory Data Analysis (Eda) And Predictive Analytics**

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements

for the Course of

DSA0404–Fundamental Of Data Science

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

S. Deva Dharshini (192472350)

Under the Supervision of

Dr. Krishnaraj Mony

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

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Saveetha Institute of Medical and Technical Sciences
Chennai – 602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Data-Driven Public Transport Delay Analysis Using Exploratory Data Analysis (EDA) and Predictive Analytics**” has been carried out by **S. Deva Dharshini (192472350)** under the supervision of **Dr.Krishnaraj Mony** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Dr.T.P Anithaashri

Professor

CSE

Saveetha School of Engineering

SIMATS, Chennai-602105

COURSE FACULTY

Dr. Krishnaraj Mony

Professor

CSE

Saveetha School of Engineering

SIMATS, Chennai-602105

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai – 602105

DECLARATION

I, **S. Deva Dharshini(192472350)** of the Department of Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**Data-Driven Public Transport Delay Analysis Using Exploratory Data Analysis (EDA) and Predictive Analytics**” represents our own academic work. The report has been prepared in accordance with engineering ethics and academic integrity. All external sources, software, datasets, analytical methods, and AI-assisted resources, where used, shall be appropriately acknowledged. We understand that any actual operational results must be supported by authentic data and reproducible analysis.

Place: Chennai

Date: 4-06-2026

Signature of the Student

S. Deva Dharshini

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

INDIVIDUAL CONTRIBUTION STATEMENT

Mandatory for team projects

Student / Register No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. Contribution (%)
B. Nikhila (192472102)	Dataset preparation and EDA	Data cleaning, feature design	EDA validation	Ch. 1–2	33%
S.Deva Dharshini(192472350)	Predictive modelling & Documentation	Model implementation	Model evaluation, Results interpretation	Ch. 3–4	34%
Y. Ramya (192424397)	Visualization and testing and integration	Dashboard/ plots&System integration	Test cases and error analysis	Ch. 45, Appendices	33%
Total					100%

ABSTRACT

Planning and managing public transportation services can be challenging for both passengers and transport operators. This project introduces a data-driven approach to examine and predict public transport delays using Exploratory Data Analysis (EDA) and predictive analytics. The study focuses on understanding how factors such as route, operating time, traffic conditions, passenger volume, weather, and day type influence transportation delays.

The proposed methodology follows a systematic sequence that includes data collection, preprocessing, exploratory analysis, feature creation, model training, performance evaluation, and result interpretation. Through EDA, important characteristics of the dataset can be identified, including data distributions, unusual observations, periods with higher delays, variations between routes, and relationships between operational factors. Predictive models can then be applied to forecast possible delays and assist transport operators in making timely decisions.

To demonstrate the proposed methodology, the report uses a clearly identified synthetic dataset along with example visualizations and calculated results. These demonstration values are intended only to illustrate the analytical process and should be replaced with genuine transportation data for actual project evaluation.

The expected outcome is a systematic and reproducible analytical framework that transforms transportation data into useful insights for improving schedules, allocating resources, and enhancing service quality. The project also addresses sustainability, ethical data usage, safety, security, risk management, and responsible handling of operational information. Overall, the study demonstrates that integrating EDA with predictive analytics can provide an effective engineering solution for identifying delay patterns and supporting informed public transportation management.

Keywords: Public Transportation, Delay Prediction, Exploratory Data Analysis, Predictive Modelling, Machine Learning, Data Analysis, Data Visualization

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
EDA	Exploratory Data Analysis	—
ML	Machine Learning	—
RMSE	Root Mean Square Error	minutes
MAE	Mean Absolute Error	minutes
R ²	Coefficient of Determination	—
CSV	Comma-Separated Values	—
API	Application Programming Interface	—
KPI	Key Performance Indicator	—
RF	Random Forest	—
XGBoost	Extreme Gradient Boosting	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Public transportation plays an important role in everyday travel, but delays can affect both passenger convenience and the efficiency of transport operations. Modern transport systems generate a large amount of operational information, including scheduled and actual arrival times, departure times, route details, stop information, passenger demand, traffic conditions, weather, and other contextual factors. When this information is properly organised and analysed, it can provide useful insights into when and where delays are most likely to occur.

Traditional analysis often focuses on reporting delays after they have happened. However, data analytics provides an opportunity to understand the reasons behind these delays and identify recurring patterns. Exploratory Data Analysis (EDA) can be used to examine variations across routes, time periods, traffic conditions, and other factors. Predictive analytics can further use these patterns to estimate the likelihood or expected level of future delays.

This project focuses on developing a data-driven analytical framework for public transport delay analysis and prediction. Each trip or stop event is treated as a data record, and the delay is calculated from scheduled and actual timestamps when such information is available. The prepared data is then explored using statistical analysis and visualization techniques before applying predictive models. The resulting insights can help identify high-delay periods, frequently affected routes, and operational conditions that may contribute to delays.

1.2 Need for the Project

Public transport delays are influenced by several factors and can vary significantly across routes and time periods. A systematic analytical approach is therefore required to understand these variations and support better planning.

- Delays can increase passenger waiting time and make journey durations less predictable.
- Transport operators need reliable information to identify routes and time periods that experience repeated delays.
- Large transportation datasets are difficult to examine manually and may contain relationships that are not immediately visible.

- Exploratory Data Analysis helps reveal important trends, patterns, distributions, correlations, and unusual observations within the data.
- Predictive analytics can provide advance estimates of expected delays and support timely operational decisions.

The main engineering need is to bring data preparation, analysis, visualization, predictive modelling, and interpretation together into a single systematic workflow. Such a framework can convert raw transportation records into useful information for improving service planning and operational efficiency.

1.3 Problem Statement

Public transport delays are affected by multiple operational and environmental factors, making them difficult to analyse using simple statistical summaries alone. The project therefore aims to develop a data-driven system that can identify important delay patterns through Exploratory Data Analysis and predict expected transport delays using suitable machine-learning techniques. The system should also consider data quality, model interpretability, computational requirements, privacy, security, sustainability, and practical operational limitations.

1.4 Project Aim

The aim of this project is to develop a reproducible data analytics framework for analysing public transport delays and predicting expected delay levels using Exploratory Data Analysis and machine-learning techniques. The framework is intended to provide clear and interpretable insights that can assist in transport planning and operational decision-making.

1.5 Project Objectives

1. To prepare a structured dataset containing public transport delay information and relevant operational and environmental factors.
2. To clean, transform, and validate the collected data to ensure its suitability for analysis.
3. To investigate delay patterns using statistical summaries, visualizations, time-based analysis, route comparisons, and relationship analysis.
4. To identify and construct relevant features such as peak-hour indicators, traffic categories, and encoded categorical variables.
5. To develop baseline and machine-learning models for predicting public transport delays.

6. To compare model performance using evaluation measures such as MAE, RMSE, and R^2 .
7. To interpret the analytical and modelling results and determine their usefulness for public transport planning.
8. To examine the limitations, risks, ethical considerations, and possible improvements associated with the proposed system.

1.6 Scope

The project covers the preparation and analysis of public transport data, including delay calculation where scheduled and actual timestamps are available. It includes data cleaning, missing-value treatment, outlier identification, descriptive statistics, route-wise and time-wise analysis, weather and traffic analysis, data visualization, feature engineering, predictive modelling, model evaluation, and interpretation of results.

The project does not involve direct control of buses, traffic signals, or other physical transportation infrastructure. Passenger payment or ticketing transactions and individual passenger profiling are also outside the scope of the study. Real-time deployment is not included in the current implementation unless a reliable and authenticated live-data source is available.

The analysis assumes that the available data contains sufficiently consistent timestamps, identifiable trips or routes, meaningful delay information, and an adequate number of observations for model development and evaluation. The study is limited by the selected transport network, available data fields, analysis period, and overall quality of the dataset.

1.7 Expected Engineering Outcome

The expected outcome is a functional analytical prototype that demonstrates the complete process of public transport delay analysis and prediction. It will include a prepared dataset, data-analysis scripts or notebooks, meaningful visualizations, predictive models, model evaluation results, and an interpretation of the findings.

The proposed workflow is designed to be reproducible so that another dataset with a similar structure can be processed using the same methodology with minimal changes. This makes the framework suitable as a foundation for further development and practical transportation analytics applications.

1.8 Engineering Problem Formulation

The delay for a transport observation i can be determined from its scheduled and actual timestamps. When both values are available, the delay is defined as:

$$D_i = \max(0, T_{\text{actual},i} - T_{\text{scheduled},i})$$

where D_i represents the delay, $T_{\text{actual},i}$ is the actual arrival or departure time, and $T_{\text{scheduled},i}$ is the corresponding scheduled time.

The prediction problem involves learning a function $f(X)$ that can estimate the expected delay based on relevant input variables. These variables may include route, operating hour, day type, traffic conditions, passenger load, weather conditions, and other available operational information.

The proposed system must not focus only on prediction accuracy. It should also provide results that are understandable, computationally practical, reasonably robust to data-quality issues, and useful for real-world transportation planning and decision-making.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review focuses on four major areas related to the proposed system: methods for measuring public transport delays, the use of Exploratory Data Analysis in transportation datasets, machine-learning techniques for delay prediction, and visualization methods that support operational decision-making. Different approaches are considered based on their ability to analyse transport behaviour, predict delays, and provide results that are understandable and useful to users.

The review also considers the balance between model accuracy and complexity. While advanced models may provide better predictions in some situations, simpler models can be easier to understand and implement. Therefore, the selection of techniques for this project is based not only on predictive performance but also on interpretability and practical applicability.

For the final project report, the review can be further supported with the research papers, datasets, technical standards, and software tools that are actually used during the implementation. The discussion presented here provides an engineering-level overview of the approaches relevant to the proposed system rather than a complete systematic review of all available research.

2.2 Review of Existing Work

Earlier approaches to public transport delay analysis mainly depend on basic statistical measures such as average delay, delay frequency, punctuality rates, and route-wise performance reports. These methods are straightforward and useful for presenting historical performance, but they may not fully capture the combined influence of factors such as traffic, passenger demand, weather, route characteristics, and operating time.

Exploratory Data Analysis provides a more detailed way of studying transportation data. It allows researchers to examine the overall structure of the dataset, identify missing or unusual observations, study delay distributions, compare routes, and analyse changes across different time periods. Performing this analysis before model development is important because problems such as incorrect data, extreme values, or information leakage can negatively affect the reliability of predictive results.

Machine-learning approaches provide an additional layer of analysis by using historical patterns to estimate future delays. Simple regression techniques can serve as baseline models because their results are relatively easy to interpret. More advanced tree-based techniques can capture nonlinear

relationships and interactions between multiple transport-related variables. Based on the characteristics and size of the available dataset, this project therefore considers a baseline predictive model along with suitable nonlinear machine-learning models.

Overall, existing approaches show that combining descriptive analysis with predictive modelling can provide a more complete understanding of public transport delays. This forms the basis for the proposed framework, which integrates data preparation, EDA, feature engineering, predictive modelling, evaluation, and interpretation into a single workflow.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual/historical reporting	Simple; easy to understand	Limited multivariable analysis; retrospective	Used as baseline reporting layer
Spreadsheet-based EDA	Low setup cost; familiar tools	Harder to reproduce at scale; manual steps	Useful for early inspection
Statistical regression	Interpretable; efficient	May miss nonlinear interactions	Baseline predictive model
Tree-based machine learning	Handles nonlinear relationships and mixed features	Less transparent; requires validation	Candidate predictive approach
Real-time prediction platform	Potentially proactive	Higher integration, data-quality and infrastructure requirements	Future deployment direction

2.4 Research / Engineering Gap

Existing public transport studies often focus either on analysing historical delay patterns or on developing prediction models. However, there is a need for a more complete approach that connects data preparation, quality checking, exploratory analysis, predictive modelling, model evaluation, and practical interpretation within a single workflow. A highly accurate prediction model may still produce unreliable results when the input data contains errors, missing information, or inconsistent

records. Similarly, predictions have limited practical value if transport operators cannot understand the factors influencing them.

The proposed project addresses this gap by bringing these different stages together into one structured analytical framework. The framework is developed at the prototype level and is intended to demonstrate how transport data can be transformed into reliable and understandable information for decision-making.

2.5 Proposed Contribution

The main contribution of this project is the development of an end-to-end framework for public transport delay analysis and prediction. The proposed workflow begins with defining and calculating delay, followed by data validation, preprocessing, Exploratory Data Analysis, feature engineering, predictive model development, and performance evaluation.

The framework also focuses on interpreting the results rather than reporting prediction accuracy alone. Important variables and delay patterns are examined to identify insights that may be useful for transport planning and operational improvement. Since the workflow follows a structured and reproducible process, it can be adapted to other public transport datasets with similar characteristics and extended with more advanced models or real-time data in future work.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Passengers	Clear identification of periods and routes associated with higher delays.
Transport operators	Actionable evidence about recurring delay patterns.
Analyst / project team	Reproducible data-processing and modelling workflow.
Management	Concise KPIs, comparisons, and evidence for planning decisions.
Institution / evaluator	Traceable methodology, testing, limitations, and ethical handling of data.

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Import structured transport records	Functional	Dataset loads without schema errors
Validate and clean data	Functional	Missing/invalid records identified and documented
Generate EDA summaries and plots	Functional	Key variables covered by tables/figures
Train predictive models	Functional	At least one baseline and one candidate model
Evaluate predictions	Functional	MAE, RMSE and R^2 reported where appropriate

Reproducibility	Non-Functional	Same input and code produce same outputs
Interpretability	Non-Functional	Important variables and limitations documented
Privacy/security	Non-Functional	No unnecessary passenger identifiers used

3.2 Constraints & Applicable Standards

Only directly applicable standards and institutional requirements should be listed in the final submission. The project primarily depends on data governance, software reproducibility, and academic integrity rather than physical hardware standards.

Standard / Code	Requirement	How Applied in This Project
Institutional academic integrity policy	Accurate attribution and reporting	Sources, datasets, tools, and AI-assisted resources acknowledged
Dataset provider terms / licence	Use data only within permitted conditions	Licence and access conditions recorded in project documentation
Applicable privacy/data-protection requirements	Avoid unnecessary personal data	Passenger-level identifiers excluded or anonymised where applicable
IEEE-style referencing	Consistent technical citation	References formatted consistently in final report

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Delay definition	Consistent timestamp-based definition	minutes	High

Data completeness	Missingness quantified before modelling	%	High
Model evaluation	Use held-out test data	dataset split	High
Prediction error	Minimise MAE/RMSE subject to interpretability	minutes	High
Processing time	Practical for project-scale datasets	seconds/minutes	Medium
Reproducibility	Fixed preprocessing/model configuration	—	High
Privacy	No unnecessary personally identifiable information	—	High

3.4 Alternative Solutions & Evaluation

Alternative 1: A descriptive EDA-only solution. It provides strong interpretability and low computational cost but cannot directly estimate future or unseen delay.

Alternative 2: A regression baseline plus a tree-based predictive model. It combines an interpretable reference point with the ability to capture nonlinear relationships.

Alternative 3: A real-time streaming prediction system. It could provide operational alerts but requires reliable live data feeds, monitoring, deployment infrastructure, and stronger validation.

Criterion	Weight	Alternative 1 EDA Only	Alternative 2 EDA + ML	Alternative 3 Real-Time
Performance	30%	3/5	5/5	5/5
Cost	20%	5/5	4/5	2/5
Safety	15%	4/5	4/5	3/5
Sustainability	15%	5/5	4/5	3/5
Reliability	20%	4/5	4/5	3/5

3.5 Selected Design & Detailed Design

Alternative 2 is selected because it provides a balanced engineering solution for a capstone project: EDA establishes data quality and interpretable patterns, while predictive modelling extends the system toward proactive estimation. It avoids the infrastructure burden of a real-time platform while still demonstrating meaningful analytics and model evaluation.



Figure 3.1. Proposed analytical system architecture

The system is organised as a sequence of dependent stages. Data acquisition supplies the raw records; cleaning and validation ensure the data are suitable for analysis; EDA identifies patterns and anomalies; feature engineering converts raw variables into model-ready predictors; and the predictive model produces delay estimates that are evaluated and interpreted.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Model family	Linear regression vs tree-based model	Linear model is transparent; tree model captures nonlinear effects	Tree model is less directly interpretable	Use linear regression as baseline and tree-based model as candidate
Validation	Random split vs time-aware split	Random split is simple; time-aware split better reflects future prediction	Time-aware split may reduce training diversity	Prefer time-aware validation when timestamps support it
Feature set	All available	All fields may	Unnecessary	Use relevant

	fields vs selected operational fields	improve fit	fields increase noise/privacy risk	operational/context variables only
Deployment	Notebook/script vs real-time service	Notebook is reproducible and low cost	No live alerts	Prototype as reproducible script/notebook;

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Prefer software analysis over additional physical infrastructure	Computational requirements and reuse of existing records	Prototype remains lightweight and reusable
Ethics	Avoid unfair or unsupported conclusions about routes or passenger groups	Feature relevance and aggregation level	Use operational variables and avoid sensitive personal attributes
Health & Safety	Predictions should support planning, not autonomous safety-critical control	Model uncertainty and error analysis	Human review required for operational decisions
Security	Protect operational datasets and access credentials	Data fields, storage, access controls	Exclude unnecessary identifiers and secrets

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Missing or inconsistent	High	High	High	Validate schema; quantify	Medium

timestamps				missingness; document exclusions	
Data leakage into model	Medium	High	High	Use temporal/appropriate train-test separation and feature review	Low- Medium
Model overfitting	Medium	High	High	Cross-validation and held-out testing	Low- Medium
Biased or incomplete route coverage	Medium	Medium	Medium	Report coverage limitations and avoid overgeneralisation	Medium
Unauthorised data access	Low- Medium	High	Medium	Access control; remove unnecessary identifiers	Low
Misinterpretation of predictions	Medium	High	High	Report uncertainty/error and require human judgement	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The implementation follows an iterative data-science workflow. First, the dataset schema and delay definition are established. Next, records are cleaned and inspected through EDA. After exploratory findings are documented, features are prepared for modelling and baseline/candidate models are trained. Finally, the models are evaluated on held-out data and the results are interpreted against the original engineering objectives.

Resource	Purpose	Stage Used
Python	Data processing and modelling	All stages
Pandas	Tabular data cleaning and transformation	Data preparation / EDA
NumPy	Numerical computation	Feature preparation
Matplotlib	Black-and-white analytical plots	EDA / results
Scikit-learn	Model training and evaluation	Predictive analytics
Jupyter Notebook / Python script	Reproducible workflow	Implementation and testing
CSV/Excel transport records	Input data format	Data acquisition

4.2 Software Implementation & Modern Tools Used

The prototype can be implemented as a Python notebook or modular Python script. The recommended modules are: data loading, validation, preprocessing, EDA, feature engineering, model training, evaluation, and reporting. Only the tools actually used in the final project should remain in the submitted tool list.

A reproducible environment should record Python version, package versions, random seeds where applicable, dataset version, and the exact model configuration. Credentials or private API keys must never be placed in the report or source code.

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Data loading	Load a representative CSV file	File loads and expected columns are present
Timestamp validity	Check parsing and impossible dates	Invalid records identified before modelling
Missing values	Compute missingness per column	Missingness reported and treatment documented
Delay calculation	Compare calculated delay with hand-checked records	Formula produces expected minutes
EDA outputs	Generate required plots/tables	All key variables represented
Model training	Train baseline and candidate model	Training completes without leakage
Model evaluation	Evaluate on held-out data	MAE/RMSE/R ² calculated where appropriate
Reproducibility	Repeat run with same configuration	Outputs remain consistent

4.4 Results

The following figures are based on a synthetic demonstration dataset of 500 transport-event records generated solely to validate the analytical workflow. They are included to show how the final report should present EDA results. They are not claimed to be measurements from a real transport operator. For an academically valid final submission, replace these demonstration results with results obtained from the project's authentic dataset and retain the same analytical structure.

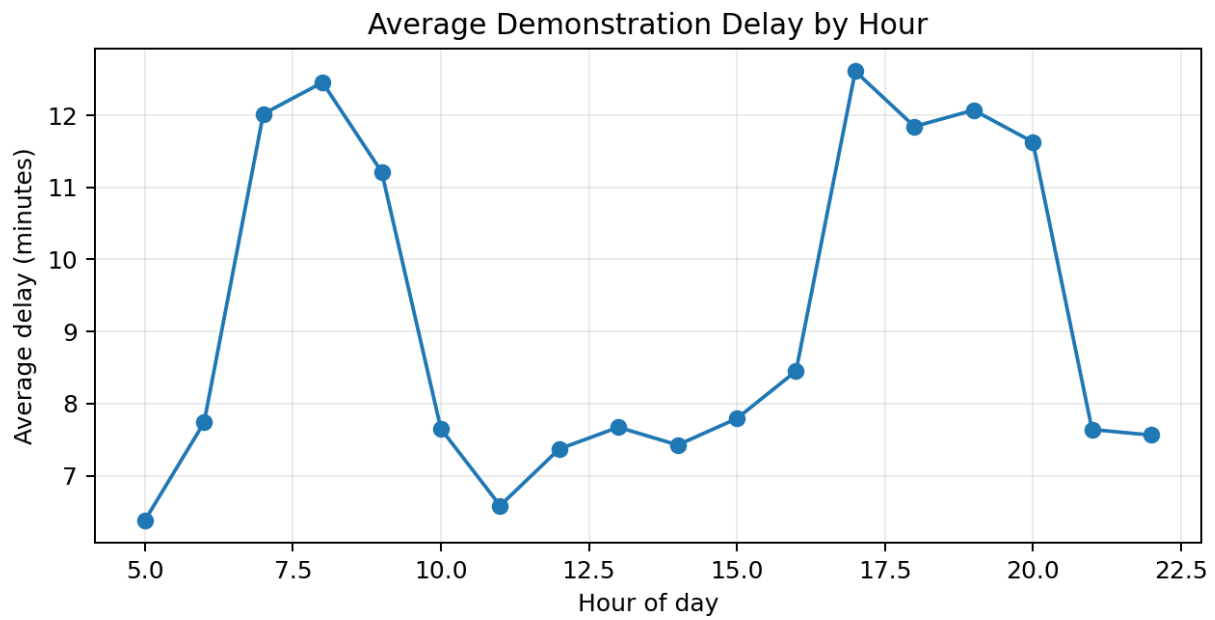


Figure 4.1. Average demonstration delay by hour

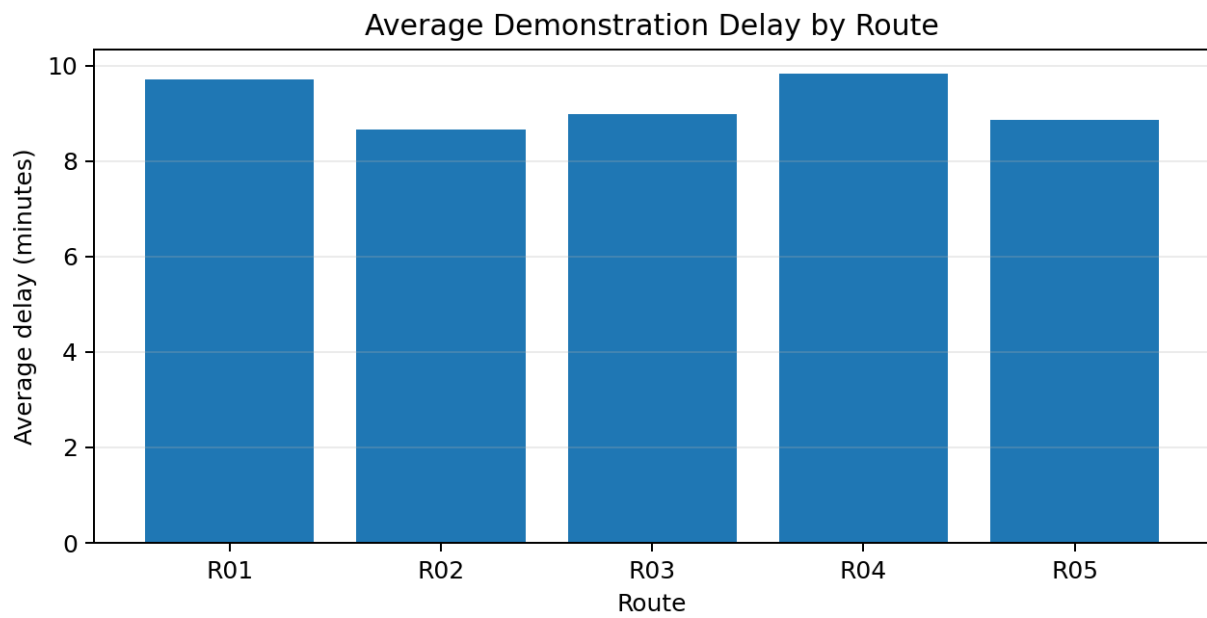


Figure 4.2. Average demonstration delay by route

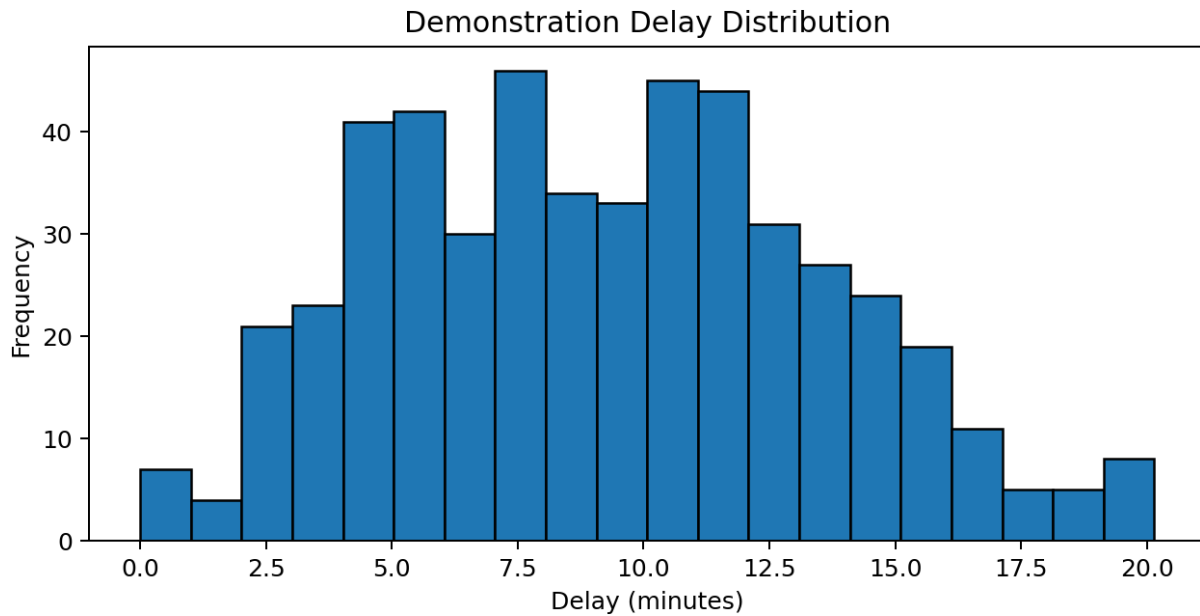


Figure 4.3. Demonstration delay distribution

Measure	Demonstration Value
Number of records	500
Mean delay	9.23 min
Median delay	9.12 min
Minimum delay	0.00 min
Maximum delay	20.15 min
Highest average-delay route	R04
Highest average-delay hour	17:00

4.5 Data Analysis & Engineering Judgment

The demonstration analysis shows the intended reasoning pattern rather than a claim about any real network. Time-of-day grouping is used to identify peak operational periods, route grouping highlights route-level differences, and the distribution plot shows whether delays are concentrated around a typical value or have a long tail. In a real dataset, these observations should be cross-checked against service schedules, traffic conditions, weather records, and data-collection coverage before an operational decision is made.

The principal engineering judgement is that EDA should precede predictive modelling. If a route contains many missing timestamps, or if a variable is only known after the delay occurs, including it

in a predictive model may produce misleading results. Similarly, a high-performing model is not automatically useful if its test set does not represent the future operating period. Model performance must therefore be considered together with data quality, validation design, interpretability, and deployment context.

Predictive Analytics Workflow



Figure 4.4. Predictive analytics workflow adapted for delay prediction

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Prepare a structured transport dataset	Schema, cleaning and validation workflow	Partially – authentic dataset must be inserted
Perform EDA	Demonstration plots and analytical plan	Fully for methodology; project data pending
Develop predictive model	Baseline + candidate model design specified	Partially – final score depends on authentic data
Evaluate model	MAE/RMSE/R ² test plan defined	Partially
Interpret engineering implications	Decision-oriented analysis included	Fully at methodology level
Document ethics, risk and security	Chapter 3.7 tables	Fully

4.7 Strengths & Limitations

Strengths

- Brings together Exploratory Data Analysis and predictive modelling to provide a more complete understanding of transport delays.
- Follows a structured and repeatable workflow that can be applied to different transportation datasets.
- Considers important factors such as data validation, risk management, ethical practices, privacy, and model interpretation.
- Enables analysis of delay patterns based on specific routes and different time periods.
- Provides a foundation for future real-time delay prediction when suitable live transportation data become available.

Limitations

- The accuracy of the models depends on the quality, availability, and completeness of the historical transportation data.
- Changes in routes, schedules, traffic patterns, or operational policies may reduce the accuracy of models trained on past data.
- Predicted delay values represent estimates and should not be considered exact or guaranteed outcomes.
- Real-time implementation would require continuous data collection, system monitoring, model updates, and validation under actual operating conditions.
- The results presented in this report are based on synthetic data and therefore cannot be treated as actual real-world performance measurements.

4.8 Project Planning, Roles & Budget

Milestone	Planned Activity	Output
Week 1	Problem definition and data schema	Problem statement + schema

Week 2	Data collection / construction	Raw dataset
Week 3	Cleaning and validation	Clean dataset
Week 4	EDA	Tables and figures
Week 5	Feature engineering	Model-ready dataset
Week 6	Predictive modelling	Baseline and candidate models
Week 7	Testing and interpretation	Evaluation results
Week 8	Documentation and presentation	Final report + viva material

Item	Estimated Cost (INR)	Actual Cost (INR)
Computing / existing laboratory systems	0–2000	To be updated
Dataset / API access, if applicable	0–3000	To be updated
Printing / binding	500–1500	To be updated
Miscellaneous	500	To be updated
Total	1000–7000	To be updated

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project develops a data-driven approach for analysing and predicting delays in public transportation using Exploratory Data Analysis and predictive analytics. The proposed approach focuses on defining a reliable delay measure, checking the quality of transportation data, identifying patterns across different time periods and routes, creating relevant predictive features, and assessing suitable models using quantitative performance measures. The study also considers important aspects such as design decisions, sustainability, ethical considerations, safety, security, risk management, testing, project planning, and system limitations.

The main contribution of the project is the combination of EDA and predictive modelling within a structured and reproducible workflow. EDA helps identify data quality issues and understand the major patterns associated with transport delays, while predictive modelling supports the estimation of potential delays in advance. The synthetic dataset used in Chapter 4 demonstrates the proposed analysis and reporting process. However, actual transportation data will be required to validate the system and make reliable conclusions regarding real-world model performance.

5.2 Future Work

- Incorporate live vehicle tracking and real-time service data to provide more accurate delay predictions.
- Experiment with advanced machine-learning algorithms, including Random Forest, Gradient Boosting, XGBoost, and suitable time-series models.
- Implement time-based validation techniques and model-drift detection to maintain prediction accuracy as traffic conditions and transport schedules change.
- Create an interactive visualization dashboard that allows users to analyze routes, travel times, and potential delay risks.
- Apply explainable AI techniques to identify and present the key factors influencing delay predictions.
- Study the impact of operational strategies, such as modifying schedules, reallocating vehicles, and optimizing routes, on overall service performance.
- Introduce prediction uncertainty and confidence ranges to provide users with more reliable information instead of relying only on single-value predictions.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Data cleaning and validation for real-world tabular records.
- Exploratory visualization and interpretation of operational data.
- Feature engineering and supervised predictive modelling.
- Model evaluation using error and goodness-of-fit metrics.
- Responsible handling of operational data and reproducible analytics.

Engineering & Professional Skills Developed

- Problem formulation and requirements analysis.
- Python-based data analysis and technical documentation.
- Experimental design and evidence-based engineering judgement.
- Team coordination, task allocation, and progress tracking.
- Technical communication through tables, figures, and concise conclusions.

Individual Reflection –B.Nikhila

The most difficult engineering task was expected to be converting raw transport records into a consistent delay variable. I addressed this by defining the timestamp logic before modelling and validating representative records manually. A key decision was to place EDA before predictive modelling because data quality can directly affect model validity. I also learned how feature choices influence both predictive performance and interpretability. If the project were repeated, I would plan earlier for time-aware validation and real operational data.

Individual Reflection – S.Deva Dharshini

The major learning outcome was understanding how a baseline model can be used to judge whether a more complex model provides meaningful improvement. I focused on model training and evaluation and learned the importance of separating training information from test information. The main engineering decision was to compare model error using more than one metric. In a future version, I would explore model explainability and drift monitoring.

Individual Reflection –Y.Ramya

My main contribution was integrating the technical work into a structured engineering report. I learned that a strong project report must explain not only what was built but also why a design was

selected, how it was tested, and what its limitations are. I also developed awareness of ethics, security, and academic integrity in data projects. If repeated, I would maintain a more detailed project log from the first week.

REFERENCES

1. W. McKinney, Python for Data Analysis, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022.
2. T. Hastie, R. Tibshirani, and J. Friedman, The Elements of Statistical Learning, 2nd ed. New York, NY, USA: Springer, 2009.
3. C. M. Bishop, Pattern Recognition and Machine Learning. New York, NY, USA: Springer, 2006.
4. F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.
5. Python Software Foundation, Python Documentation, software documentation resource.
6. Pandas Development Team, pandas documentation, software documentation resource.
7. Matplotlib Development Team, Matplotlib documentation, software documentation resource.
8. The dataset provider or transport authority used by the project, dataset title, version/date, access details and licence.
9. Any public transport research papers, standards, websites, APIs, open-source repositories, or AI-assisted resources actually used during the project.

APPENDICES

Appendix A – Detailed Calculations

For an observation with scheduled time T_s and actual time T_a , the basic delay measure is:

$$\text{Delay} = \max(0, T_a - T_s)$$

For model evaluation, Mean Absolute Error (MAE) is calculated as the mean absolute difference between actual and predicted delay. Root Mean Square Error (RMSE) gives greater weight to larger errors. R^2 indicates the proportion of variance explained by the model relative to the chosen baseline.

Metric	Expression	Interpretation
MAE	$\text{mean}(y - \hat{y})$	Average absolute prediction error
RMSE	$\text{sqrt}(\text{mean}((y - \hat{y})^2))$	Penalises larger errors more strongly
R^2	$1 - \text{SSE}/\text{SST}$	Relative explanatory performance

Appendix B – Engineering Drawings / Schematics

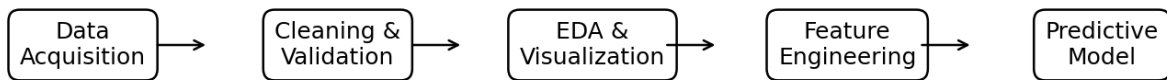


Figure B.1. System architecture for the public transport delay analytics prototype

Appendix C – Source Code / Code Repository Information

Only essential code excerpts should normally be included in the report. The complete project source should be maintained in the approved repository. A typical processing sequence is:

- 1. Load CSV or API-derived transport records.

- 2. Parse scheduled and actual timestamps.
- 3. Calculate delay and validate impossible values.
- 4. Handle missing values and encode categorical variables.
- 5. Generate EDA tables and figures.
- 6. Split data using an appropriate validation strategy.
- 7. Train baseline and candidate predictive models.
- 8. Calculate MAE, RMSE and R^2 where appropriate.
- 9. Save plots, model configuration and evaluation results.
- 10. Generate the final analytical summary.

Appendix D – Datasheets / Data Dictionary

Field	Description	Type	Example
route_id	Route identifier	Categorical	R01
scheduled_time	Scheduled event time	Datetime	07:45
actual_time	Observed event time	Datetime	07:53
delay_min	Calculated delay	Numeric	8.0
traffic_index	Traffic intensity indicator	Numeric	65
passenger_load	Relative passenger load	Numeric	58
weather	Weather category	Categorical	Rain
day_type	Weekday/weekend indicator	Categorical	Weekday

Appendix E – Experimental / Raw Data

Attach the authentic raw-data file, data dictionary, preprocessing notes, and a record of exclusions here. The synthetic demonstration data used for the figures in this report must not be confused with operational measurements.

Appendix F – Ethics / Institutional Approval

If the project uses public or anonymised transport data without personal information, document the data source, licence, privacy treatment, and institutional requirements. If identifiable passenger

information or restricted operational data are introduced, obtain the required approvals before analysis.

Appendix H – User / Industry Feedback

Record feedback from a transport operator, faculty reviewer, or other relevant stakeholder where available. Include the date, role of the reviewer, key comments, and resulting design changes.

Appendix I – Publications / Patents / Competitions / Product Outcomes

If applicable, list conference submissions, publications, competitions, demonstrations, dashboards, or other project outcomes. If none, state “Not applicable.”

Appendix J – Individual Contribution Evidence

Include repository commit summaries, meeting records, task allocation, screenshots of individual work, or other evidence supporting the contribution percentages stated in the Individual Contribution Statement.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3.7
Teamwork and leadership	SO5	Ch. 4.8
Experimentation, analysis, engineering judgment	SO6	Ch. 4.3–4.6
Independent acquisition of new knowledge	SO7	Ch. 5.3
Standards	SO2	Ch. 3.2
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3.7
Risk assessment and mitigation	SO2/SO4	Ch. 3.7
Security considerations	Relevant SO	Ch. 3.7
Integrated/system-level engineering	SO1/SO2	Ch. 3.5
Project planning and management	SO5	Ch. 4.8
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4.8